

Memory Breakdown (70B model, 4K context)

Model Weights
140 GB (FP16)

KV Cache
~5 GB per
batch item

Activations
~2 GB

Prefill Phase (Process entire prompt)

Input Prompt
(N tokens)

all tokens
at once

Parallel
Attention
Compute

populate

KV Cache
(Store K,V
for all N tokens)

read all
past K,V

Decode Phase (Generate one token at a time)

New Token
(1 token)

Attend to
ALL past
K,V pairs

Next Token
Prediction

append

KV Cache
(Grows by 1
each step)